# The Preferendus Landscape

*A unified ecosystem for real-world design-decision optimisation, supported by human-in-the-loop AI-driven ODESYS-in-FIVES orchestration for better solutions with decisive impact.*

## Executive Summary

**ODESYS-in-FIVES** is a state-of-the-art multidimensional design and decision systems methodology for multi-attribute, multi-actor design optimisation and decision analysis. Structured according to the FIVES paradigm, ODESYS integrates performance, comprising **1) capability** and **2) feasibility**, preference, comprising **3) desirability** and **4) acceptability**, and **5) solvability** within a common preference-performance theoretical framework spanning the design-decision problem-solving journey from intent to impact.

ODESYS-in-FIVES uses two complementary methodologies: **IMAP** for multi-objective design optimisation (MODO), supported by the **IGS solver family**, and **TOMAP** for multi-criteria decision analysis (MCDA). Performance information can be represented across four levels of modelling complexity, ranging from directly assigned scores (Level 0) to advanced simulation models (Level 3). At the core of both methodologies, the **A-FINE Aggregator** derives decision-valid aggregated preferences within a unified preference-performance domain.

The methods and algorithms are operationalised within the **Preferendus software suite**, with Preferendus-Agent as its central architecture. **Preferendus-Agent** provides an AI-driven orchestration layer for formulating and solving Problems of Interest (POIs). Both a Dialogue Agent and a Delivery Agent are equipped with ODESYS-in-FIVES skills, enabling them to work natively with Level 0 and Level 1 performance representations while seamlessly connecting to real-world-specific Level 2 and Level 3 performance models.

## Architecture Overview

| Layer | Method / Technology | Preferendus Product | Role |
|---|---|---|---|
| Design & Decision Systems | ODESYS-in-FIVES | Preferendus-Agent | Orchestrates ODESYS methods, including IMAP and TOMAP, and their capabilities within a unified AI-driven design and decision environment. It natively supports performance Levels 0 and 1 and can connect to enterprise-specific Level 2 and Level 3 models. |
| Multi-Objective Design Optimization (MODO) | IMAP: Integrative Maximization of Aggregated Preferences | Preferendus-IMAP-IGS | Performs preference-performance-driven design optimization through IMAP, the IGS solver family, and A-FINE. |
| Best-Fit Candidate Search | IGS Solver Family | Preferendus-IMAP-IGS | Searches complex design spaces for best-fit-for-common-purpose solutions. |
| Multi-Criteria Decision Analysis (MCDA) | TOMAP: Trade-Off Matrix by Aggregated Preferences | Preferendus-TOMAP | Produces preference-based trade-off matrices, rankings, analyses, and design-decision support. |
| Preference Aggregation | A-FINE Aggregator, the core aggregation algorithm | Preferendus-ALL | Derives decision-valid aggregated preferences in the Linear Preference Space (LPS). |

## ODESYS Methodology Stack

- **A-FINE Aggregator:** Core preference-aggregation algorithm that maps commensurable preference information to a decision-valid weighted centroid in the LPS.
- **IGS Solver Family:** Family of intergenerational solvers for best-fit candidate search in complex MODO design spaces.
- **IMAP:** Integrative Maximization of Aggregated Preferences, the ODESYS methodology for preference-performance-driven MODO.
- **TOMAP:** Trade-Off Matrix by Aggregated Preferences, the ODESYS methodology for preference-based MCDA.
- **ODESYS-in-FIVES:** The skills-based framework that structures performance, preference, and solvability and enables AI-assisted design and decision support.

## Preferendus Software Suite

**Preferendus-IMAP-IGS**
MODO-IMAP solver-family platform implementing IMAP through the IGS solver family and the A-FINE Aggregator for preference-performance-driven design optimisation.

**Preferendus-TOMAP**
MCDA-TOMAP tool implementing TOMAP and the A-FINE Aggregator for preference-based trade-off matrices, ranking, trade-off analysis, and design-decision support.

**Preferendus-Agent**
AI-driven design and decision agent trained with ODESYS-in-FIVES skills and orchestrating IMAP, TOMAP, A-FINE, and IGS capabilities within a unified environment.

## Performance Modelling Levels

Performance information can be represented within Preferendus-Agent at four levels. These levels distinguish how performance information is represented, the complexity of the underlying model, and how the information is made available to the design and decision process. Level 0 uses discrete raw performance values for a set of predefined, feasible, and acceptable alternatives. Levels 1 to 3 provide performance functions or models that generate performance values for variable design inputs, using generic analytical equations, enterprise-specific adaptations or surrogate models, and advanced behavioural or simulation models, respectively.
Preferendus-Agent natively supports Level 0 and Level 1 performance representations, making Preferendus-Agent directly applicable to a wide range of design and decision problems. Enterprise-specific Level 2 and Level 3 capabilities can be made available to Preferendus-Agent through dedicated AI skills, tools, model interfaces, and software connectors. Preferendus-Agent orchestrates these levels and transforms their performance outputs into preference-meaningful and decision-valid design optimisation and decision support.

### Level 0: Discrete Raw Performance Values

Discrete performance values are available for a finite set of predefined, feasible, and acceptable alternatives. These values may originate from measurements, calculations, expert judgement, historical data, or external models. Within Preferendus-Agent, they are treated as given values and evaluated through preference functions.

**Examples**: cost, time, weight, capacity, emissions, reliability, catalogue specifications, or previously calculated results.

## Level 1: Generic Analytical Equations

Transparent and generally applicable heuristics and mathematical, physical, engineering, economic, or social-science equations that calculate performance as a function of design-decision parameters. These models enable Preferendus-Agent to evaluate performance beyond a predefined set of alternatives.

**Examples**: calculation rules, engineering formulas, cost functions, or physical relationships.

## Level 2: Enterprise-Specific and Surrogate Models

Calibrated, empirical, measurement-based, statistical, reduced-order, or machine-learning models represent enterprise-specific performance. They may adapt generic Level 1 equations or approximate detailed Level 3 simulations.

**Examples**: enterprise-specific performance curves, regression models, response surfaces, Gaussian processes, reduced-order models, or trained machine-learning models.

## Level 3: Advanced Behavioral and Simulation Models

High-fidelity models and specialist software calculate performance by representing detailed physical, dynamic, operational, or organisational system behaviour. They may be connected directly to Preferendus-Agent or used to generate data for Level 2 models.

**Examples**: discrete-event simulation (DES) models, graph-based models, digital twin models, and spatially discretised simulation models, such as Finite Element Method (FEM) models and Computational Fluid Dynamics (CFD) models, as well as other specialised enterprise performance software.

| Level | Representation in Preferendus-Agent | Performance Information Source |
|---|---|---|
| 0 | Fixed raw performance values for a discrete set of predefined, feasible, and acceptable alternatives | Measurements, calculations, catalogues, expert estimates, historical data, or outputs from external models |
| 1 | Generic analytical performance functions evaluated for variable design inputs | Established mathematical, physical, engineering, economic, or social-science equations and relationships |
| 2 | Enterprise-specific adaptations, reduced-order, empirical, statistical, or surrogate performance models | Enterprise data, experiments, calibration, domain expertise, or outputs from Level 3 models |
| 3 | Detailed behavioural or simulation-based performance models evaluated for variable design inputs | Specialist engineering, simulation, operational, digital-twin, or enterprise performance software |

# Methodological Positioning

## IMAP → Preferendus-IMAP-IGS

IMAP combines A-FINE preference aggregation with the IGS solver family to identify the preferred best-fit solution within a design space.

**Comparable to:** MIP, CP, CPLEX, and evolutionary optimisation methods.

**Comparative advantage:** *For preference-performance-driven design problems, IMAP can outperform conventional mathematical programming and evolutionary optimisation approaches in identifying the best-fit-for-common-purpose solution, because it directly maximises decision-valid aggregated preferences rather than separate objectives or surrogate performance measures*.

## TOMAP → Preferendus-TOMAP

TOMAP does not search for an optimum. It directly evaluates alternatives through A-FINE aggregated preferences, revealing explicit trade-offs and producing a unique preference-based ranking for decision support.

**Comparable to:** TOPSIS, MAUT, AHP, PROMETHEE, and ELECTRE.

**Comparative advantage:** *For multi-attribute, multi-actor trade-off problems, TOMAP generates a unique, preference-consistent ranking of alternatives by directly aggregating decision-valid preferences within a unified Linear Preference Space. Unlike conventional MCDA approaches, it preserves preference meaning across criteria and actors, providing a mathematically sound basis for ranking, trade-off analysis, and decision support*.

## ODESYS-in-FIVES → Preferendus-Agent

Preferendus-Agent operationalizes the broader ODESYS methodology. As an intelligent orchestration layer, it supports the formulation, analysis, and solution of complex multi-attribute, multi-actor design and decision problems in a mathematically sound and decision-valid manner. It encompasses ODESYS methods and software tools trained in AI-skills.

**Outperforming**: AI-driven open design and decision support systems, decision-intelligence platforms, and engineering copilots.

**Comparative advantage**: *Preferendus-Agent distinguishes itself as a state-of-the-art, AI-driven open design and decision support system that keeps human judgement in the loop. Rather than producing an automated numerical ordering without clear meaning, it helps users formulate the right problem of interest and establishes the best possible synthesis between what people prefer and what the system can feasibly deliver. By orchestrating decision-valid preference aggregation, performance modelling, design optimisation, and decision analysis, Preferendus-Agent produces preference-meaningful, transparent, and mathematically sound outcomes for design and decision support*.